

Program : Diploma in Instrumentation Engineering	
Course Code : 6088	Course Title: Analytical & Biomedical Instrumentation Lab
Semester : 6	Credits: 1.5
Course Category: Program Core	
Periods per week : 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To practice quantitative and qualitative analysis of gas samples
- To familiarize with the recording of bioelectric parameters
- To study the measurement of physiological variables

Course Prerequisites:

Topic	Course code	Course name	Semester
Sample Analysis		Analytical & Biomedical Instruments	4
ECG, EEG, EMG, Blood Pressure measurement		Analytical & Biomedical Instruments	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Determine various parameters using analytical instruments.	13	Applying
CO2	Analyze different gases using analyzers.	9	Applying
CO3	Measure various physiological parameters using biomedical instruments.	12	Applying
CO4	Demonstrate different biomedical equipments.	9	Understanding
	Lab Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2		2			
CO2	3	2		2	2		
CO3	3			2		2	
CO4	3			2		2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Determine various parameters using analytical instruments.		
M1.1	Spectro photometer : Determine the concentration of given sample using Spectrophotometer.	3	Applying
M1.2	Photo Colorimeter: Determine the concentration of given sample using Photo colorimeter.	3	Applying
M1.3	Flame Photometer: Determine the concentration of given sample using Flame Photometer.	3	Applying
M1.4	pH meter: Measure the pH value of a solution using pH meter.	2	Applying
M1.5	Conductivity Meter: Measure conductivity of a solution using conductivity meter.	2	Applying
CO2	Analyze different gases using analyzers.		
M2.1	Thermal Conductivity Analyzer: Study Gas analysis by using thermal conductivity analyzer	3	Applying
M2.2	Electrical Conductivity Analyzer: Study Gas analysis by using electrical conductivity analyzer	2	Applying
M2.3	Paramagnetic O2 analyzer :Study Paramagnetic Oxygen analyzer	2	Applying
M2.4	Air Pollution Monitoring Equipment : Study air pollution monitoring equipment	2	Applying
	Lab Test – I	1	

CO3	Measure various physiological parameters using biomedical instruments.		
M3.1	Sphygmomanometer: Measure blood pressure using Sphygmomanometer	2	Applying
M3.2	Digital BP Apparatus: Measure blood pressure using Digital BP apparatus	2	Applying
M3.3	Glucometer: Measure blood sugar using glucometer	2	Applying
M3.4	Blood group: Determine blood group	2	Applying
M3.5	Infrared Thermometer: Measure body temperature using infrared thermometer	2	Applying
M3.6	Skin Contact Impedance: Measure skin contact impedance using Electrodes	2	Applying
CO4	Demonstrate different biomedical equipments		
M4.1	ECG Machine: Demonstrate ECG machine	3	Understanding
M4.2	EEG Machine: Demonstrate EEG machine	3	Understanding
M4.3	EMG Machine: Demonstrate EMG machine	9	Understanding
	Lab Test – II	1	

Text / Reference:

T/R	Book Title/Author
R1	Handbook of Analytical Instruments, R S Khandpur - Tata McGraw-Hill.
R2	Hand Book of Biomedical Instrumentation, R S Khandpur - Tata McGraw-Hill.
R3	Biomedical Instruments, Dr. Arumugham, Anuradha publications.
R4	Instrumental methods of analysis, Williard, Dean Merrit & Settle - CBS publishers.

Online Resources:

Sl. No	Website Link
1	https://www.youtube.com/watch?v=6lULYGC6lAE
2	https://www.youtube.com/watch?v=xHQM4BbR040
3	https://www.ncbi.nlm.nih.gov/books/NBK279251/
4	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4322747/
5	https://www.youtube.com/watch?v=HDyqXNHLjug
6	https://www.youtube.com/watch?v=ucWlXrWF5D8